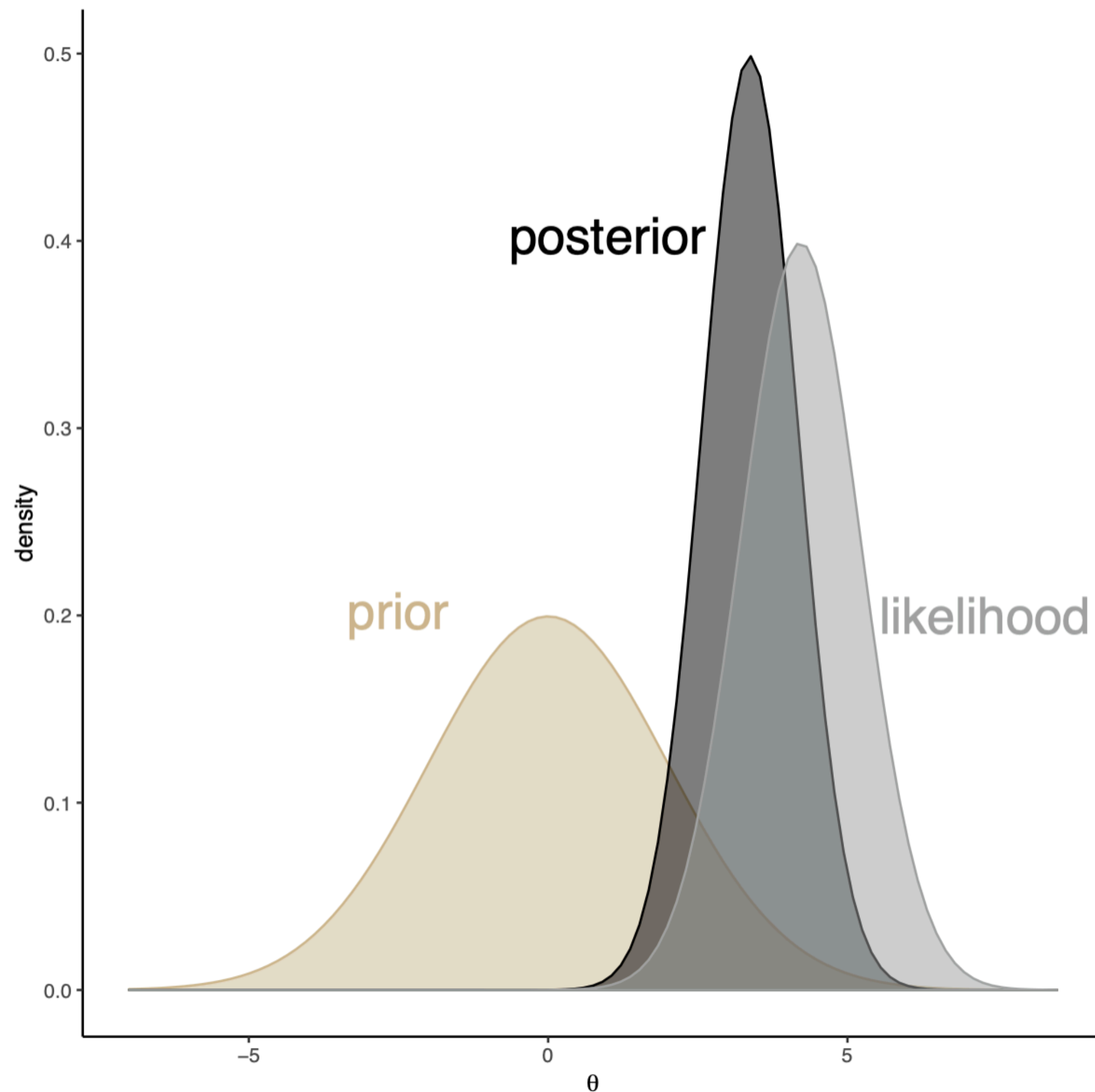
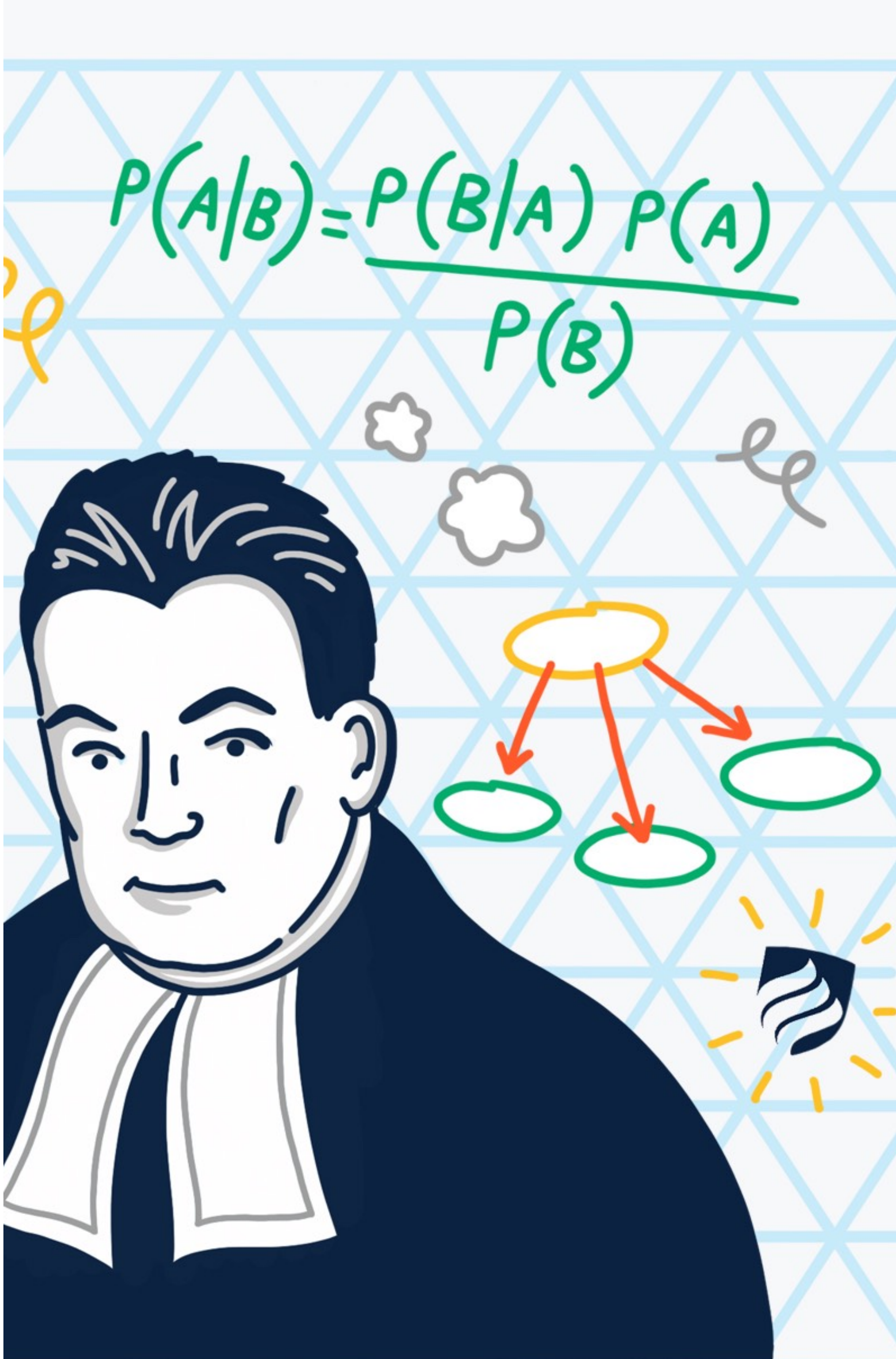




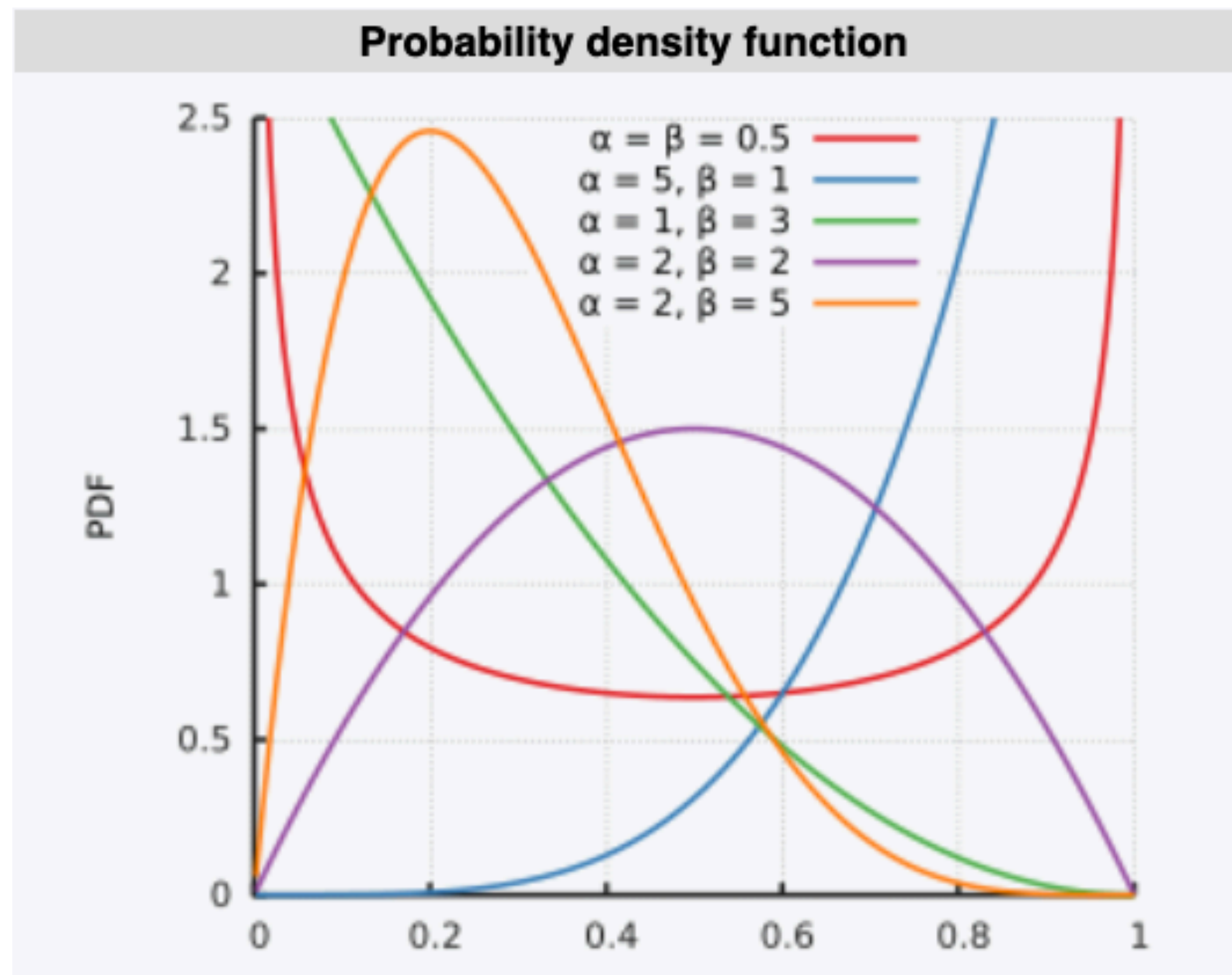
Notes on the prior distribution:

1. The prior distribution is meant to reflect our knowledge/degree of belief in the value of θ before any new data are observed.
2. It is often the case that we have prior knowledge, an expert opinion, etc. $\pi(\theta)$ is the formal way to take that knowledge into account.
3. There is a *ton* to say about what to do in the absence of prior information!
 1. "Uninformative" priors?
 2. "Objective" priors?



$$P(A/B) = \frac{P(B/A) P(A)}{P(B)}$$

From: <https://bit.ly/3gTjIBg>



Example: Binomial Bayes Estimation. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Binomial}(1, p)$. Let the prior distribution on p be a beta distribution. Find the posterior distribution of $p \mid \mathbf{x}$, i.e., the posterior distribution of the parameter p given that we observed the data $\mathbf{x}$.

- Prior distribution:
- Likelihood:
- Posterior:





Example: Binomial Bayes Estimation continued.

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Binomial}(1, p)$. Let the prior distribution on p be a beta distribution. The posterior is given by:

$$p \mid \mathbf{x} \sim \text{Beta} \left(n\bar{x} + \alpha, n - n\bar{x} + \beta \right)$$

Estimator of p ?

$$\hat{p}_1 = E(p \mid \mathbf{x}) = \frac{n\bar{x} + \alpha}{n + \alpha + \beta}$$

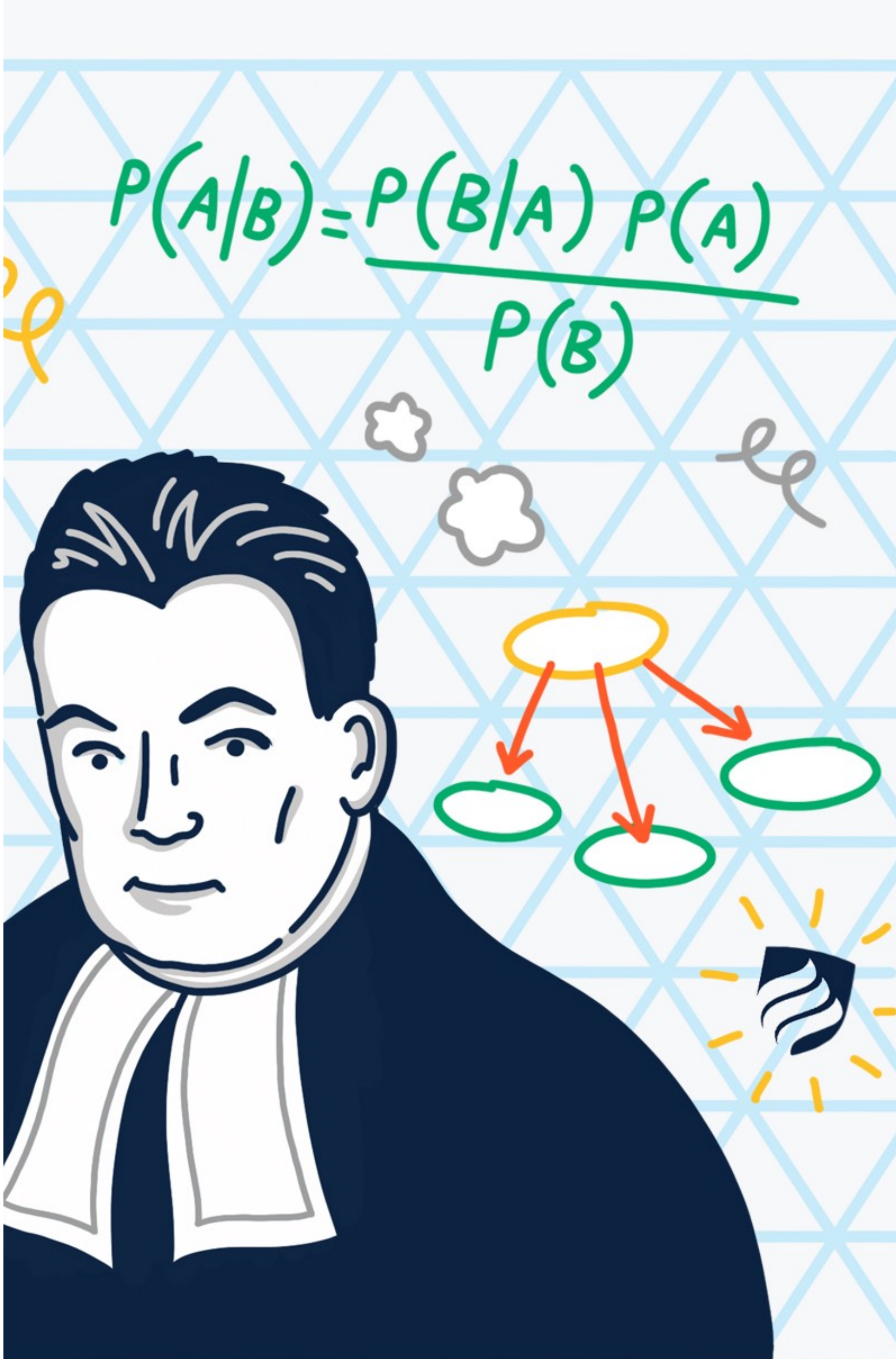


Example: Binomial Bayes Estimation continued.

Let $\hat{p}_0 = E(p) = \frac{\alpha}{\alpha + \beta}$ be the prior mean, and $\hat{p} = \bar{x}$. Then the posterior mean can also be written as:

$$E(p \mid \mathbf{x}) = \frac{n\bar{x} + \alpha}{n + \alpha + \beta} = (1 - w_n)\hat{p}_0 + w_n\hat{p},$$

$$\text{where } w_n = \frac{n}{n + \alpha + \beta}.$$



A class of prior distributions, $\mathcal{C}$, is a **conjugate family** for a likelihood, $f(\mathbf{x} \mid \theta)$, if the posterior distribution is also in class $\mathcal{C}$.

In the previous example, we saw that the beta family of distributions (prior) is a conjugate for the binomial family (likelihood).

$$a\mu^2 + b\mu + c = a\left(\mu + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)$$



Photo by Toby Elliott on [Unsplash](#)

Example: Let $X_1, \dots, X_n \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Assume σ^2 is known and we are attempting to learn about μ . Our prior is $\mu \sim N\left(\mu_0, \frac{\sigma^2}{m}\right)$. Find the posterior distribution of $\mu \mid \mathbf{x}$.

- Prior:
- Likelihood:
- Posterior:

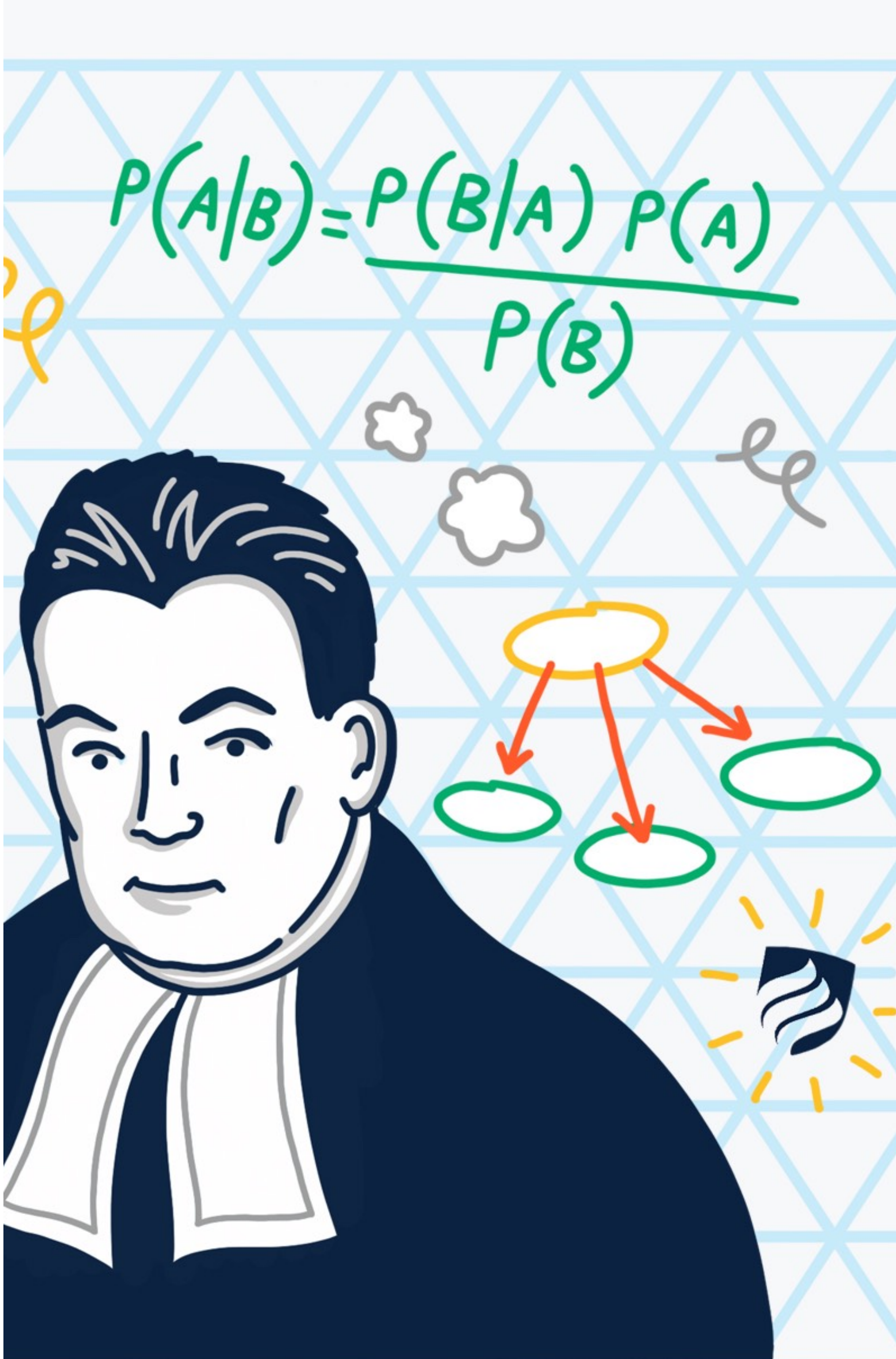


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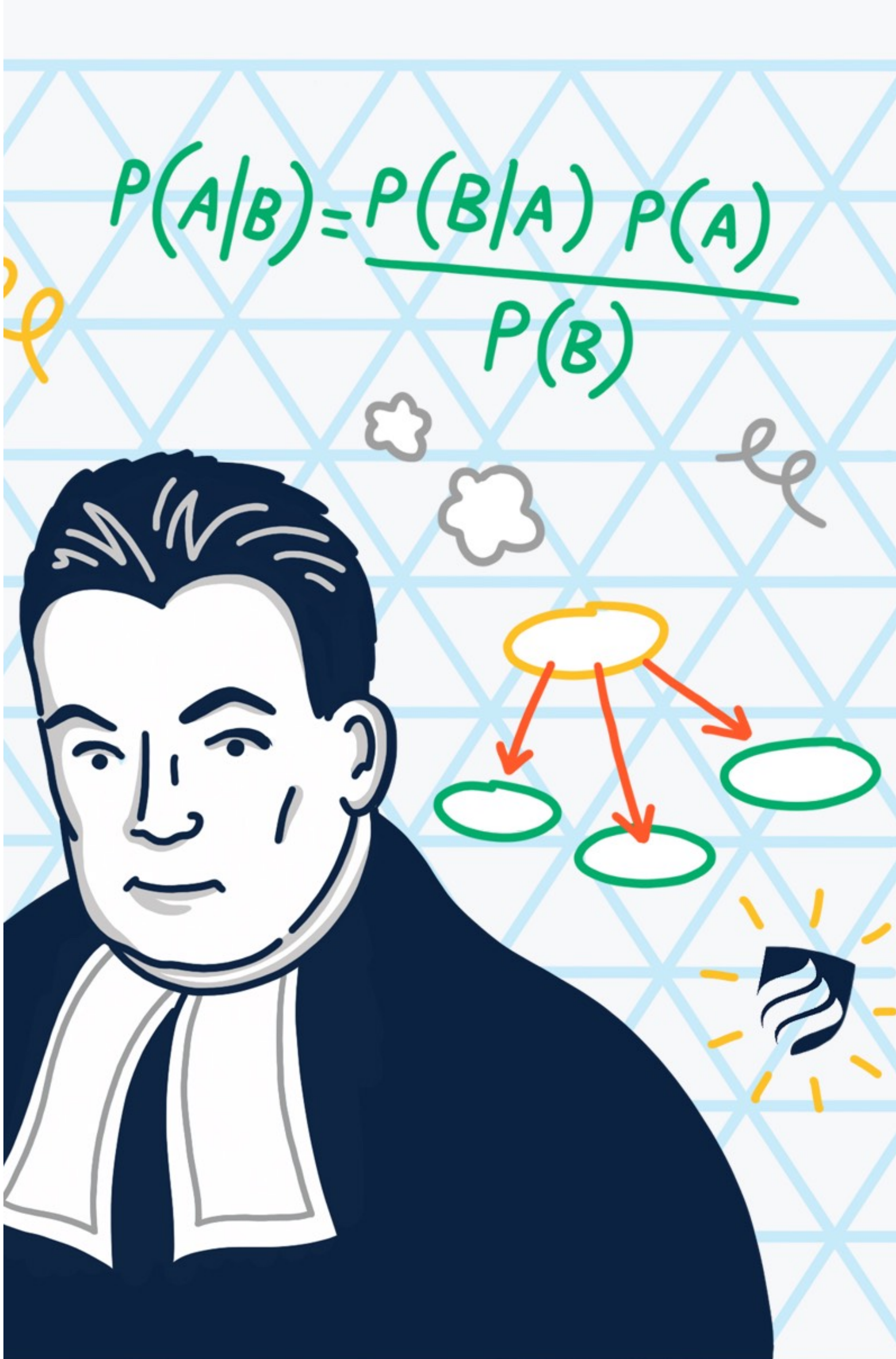
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$$\mu_1 = \frac{n\bar{x}}{n+m} + \frac{m\mu_0}{n+m} =$$

$$\sigma_1^2 = \frac{\sigma^2}{n+m} =$$

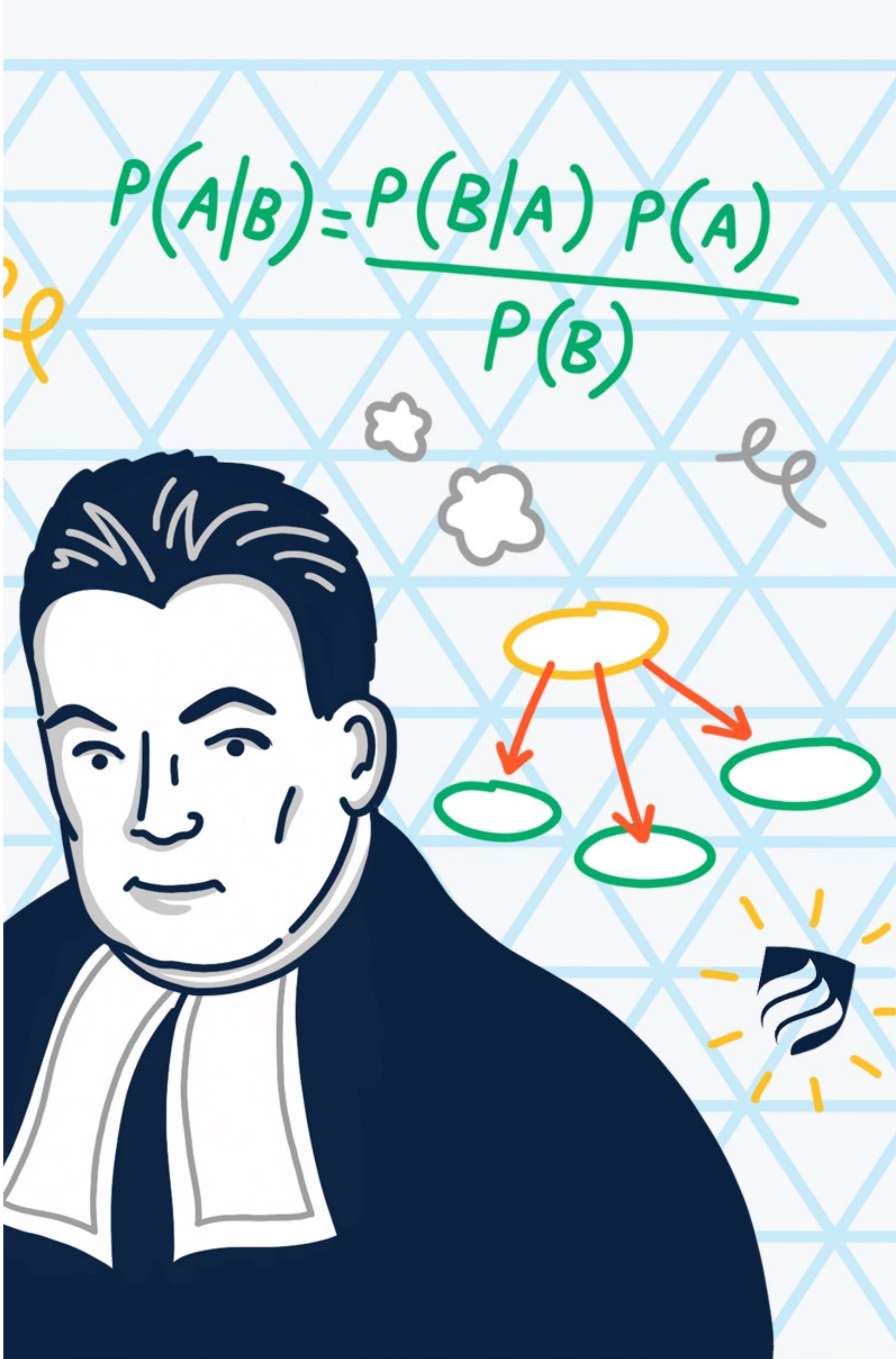


Example: Suppose researchers would like to know the probability, p , that an individual in a given population has a genetic marker that predisposes them for disease D (this genetic marker is such that an individual has it or doesn't; assume that our test for the marker is completely accurate). Researchers collected data in the following way: they tested people for this genetic marker until they found one person who had it. They stopped data collection at that point. Here's the data: $\mathbf{x} = (0,0,0,0,1)$. Suppose your prior beliefs are best represented by a $\text{beta}(2,2)$ distribution. What is the posterior distribution for $p \mid x$?



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Geometric distribution: The probability distribution of the number of failures, Y , before the first success, supported on the set $\{0,1,2,\dots\}$

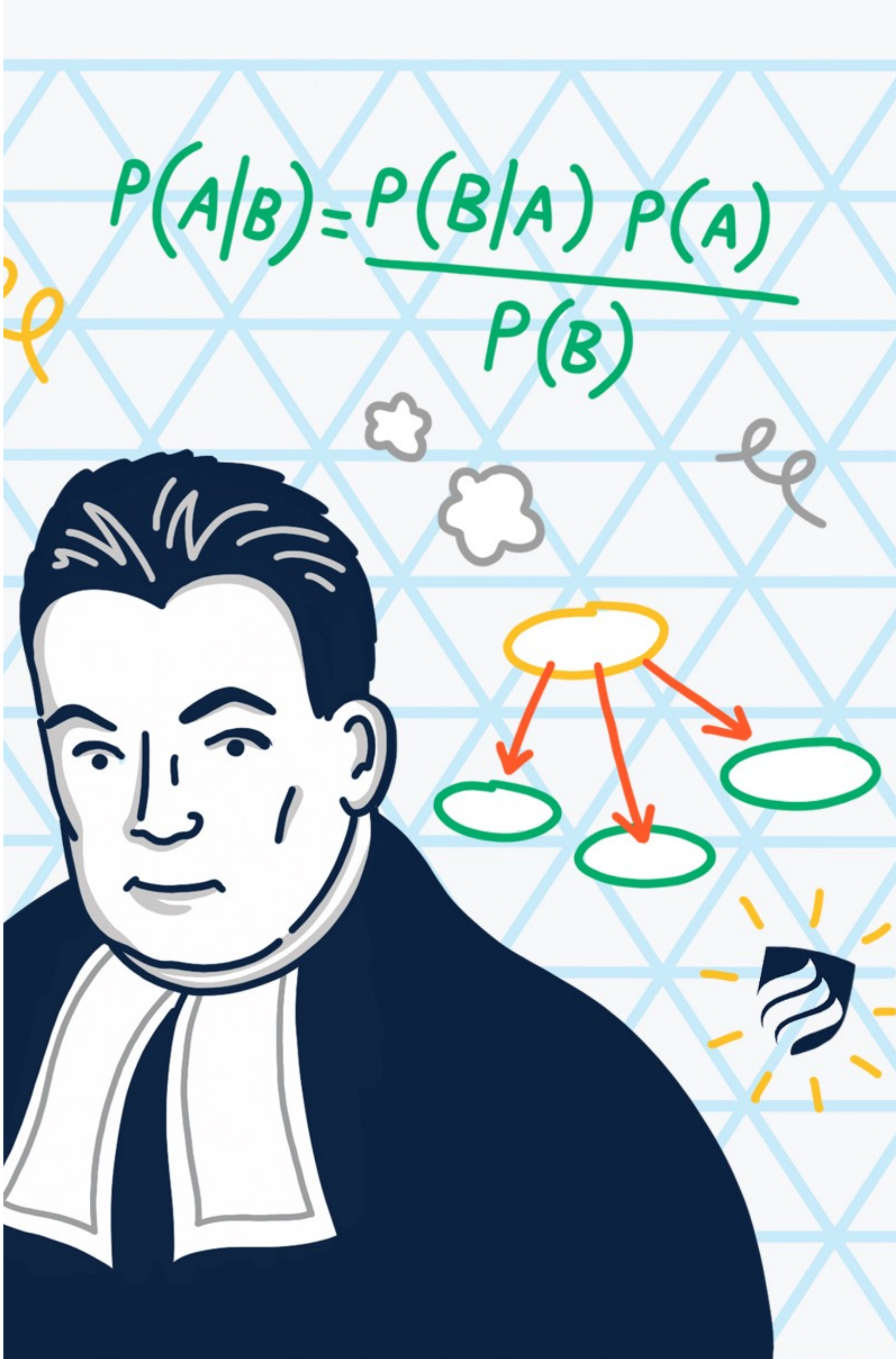


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Prior:

Likelihood:

Posterior:



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95 % credible interval? Let F^{-1} be the inverse cdf of the posterior distribution for $p \mid x$, i.e., the cdf of $\text{beta}(\alpha + 1, y + \beta)$. Then a 95 % credible interval is given by:



The inverse gamma distribution will be useful for finding the posterior of σ^2 given iid normal data. The **inverse gamma distribution** is defined as $Y = \frac{1}{X}$, where $X \sim \Gamma(a, b)$ Let's find the pdf of $Y = \frac{1}{X}$.

(Note: in general, to find the pdf of $Y = g(X)$, we can first find the cdf, and then take the derivative.)

Example: Let $X_1, \dots, X_n \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Assume μ is known and we are attempting to learn about σ^2 . Our prior is $\sigma^2 \sim \text{Inv}\Gamma(a, b)$. Find the posterior distribution of $\sigma^2 \mid \mathbf{x}$.